

Cloud as a model

Day 1 - Morning

Running scenario: Thara Hospitals patient portal pilot

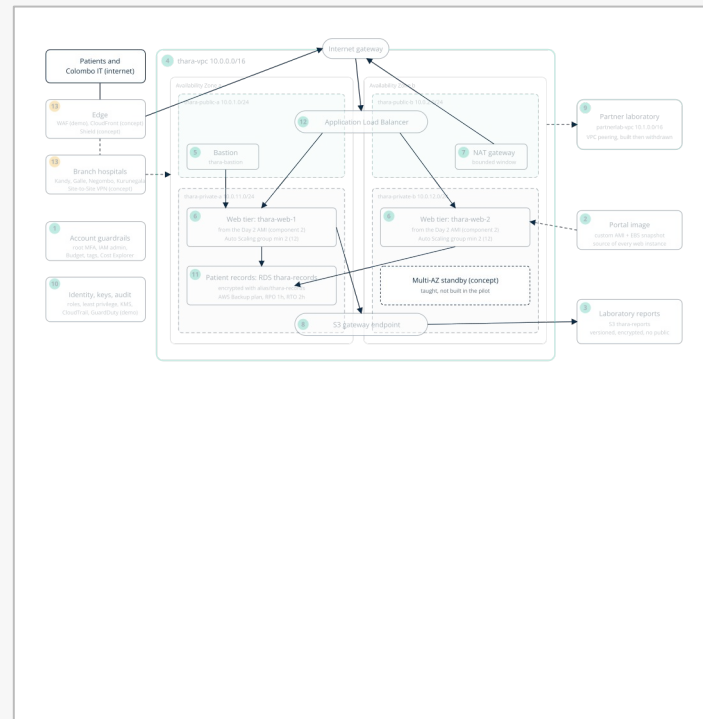
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Today

Why Thara is moving, what cloud is, where it lives, and who secures what.

- Why Thara cannot buy another server - 20 min
- What cloud is, formally - 25 min
- Where AWS is - 25 min
- Virtualisation and containerisation - 40 min
- Who secures what - 20 min



Learning outcomes

**L01**

Explain the key differences between virtualisation and containerisation.

L02

Explain cloud computing concepts.

L03

Explain high availability (HA) concepts.

L04

Describe the core features of cloud computing.

L05

Implement, scale, and manage cloud instances with storage, networking, and security features.

L06

Manage cloud data by configuring storage and backups.

L07

Explain virtual private clouds, subnets, gateways, and other cloud networking options.

L08

Describe the security features of cloud computing.

CONCEPT 1

Why Thara cannot buy another server

The monsoon flood and the CFO's USD 150 ceiling.

20 min



CFO

CHIEF FINANCIAL OFFICER

The flood cost us three days and two servers. I have approved USD 150 a month for the pilot, itemised, with the actual against the forecast every month.

The Chief Financial Officer, after the south-west monsoon flooded the basement server room.

Renting beats buying for a spiky hospital

- **Capex** buys an asset up front; the basement servers are sunk capital. **Opex** is a monthly bill for what you use
- Elasticity: the OPD peak is 07:00 to 10:00, and dengue season triples it; a bought server idles the rest of the time
- Economies of scale let a hyperscaler build what one hospital never could
- The trade-off: dependence on one provider, data in another country, new skills

YOU ALREADY KNOW THIS

Buying a core switch sized for the busiest hour of the year, with most ports dark most nights.

CONCEPT 2

What cloud is, formally

The portal is IaaS on EC2 now; a managed database is PaaS-shaped.

25 min

The five characteristics of cloud

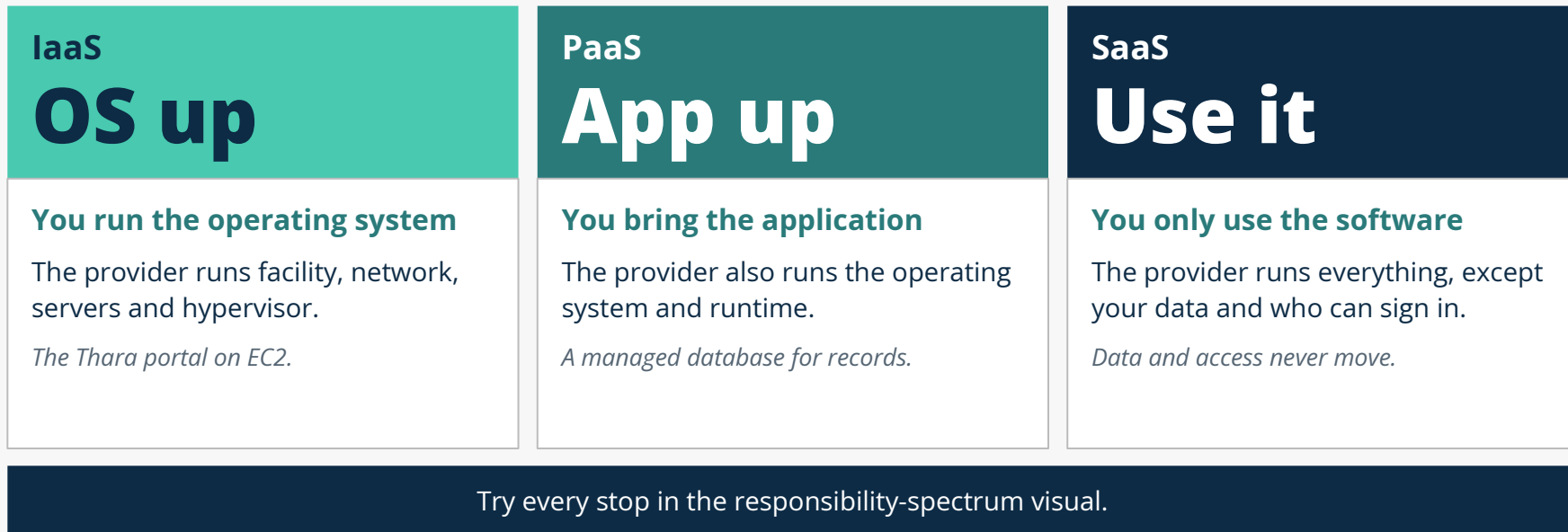


- On-demand self-service: you create resources yourself, without a ticket
- Broad network access: reached over the network from ordinary devices
- Resource pooling: many customers share the provider's hardware
- Rapid elasticity: capacity grows and shrinks quickly
- Measured service: usage is metered and billed

NIST Special Publication 800-145, The NIST Definition of Cloud Computing

If any one is missing, it is hosting, not cloud. A rented server with a yearly contract fails two of them.

Service models are a spectrum of who does what



Public, private and hybrid clouds



- **Public:** shared infrastructure run by a provider such as AWS
- **Private:** dedicated to one organisation, often in its own data centre
- **Hybrid:** the two joined; Thara's pilot is public, its future is hybrid

YOU ALREADY KNOW THIS

Leasing a rack in a colocation centre versus renting the whole service from a managed provider.

CONCEPT 3



Where AWS is

Colombo has one basement; Mumbai has three availability zones.

25 min

Regions, availability zones and edge locations

- A **region** is a geographic area; an **AZ** is one or more data centres with its own power and network
- AZs exist so one building can fail without the service failing: the first seed of high availability
- Edge locations cache content near users; they never run your servers
- IAM is global and has no region selector; EC2 is regional

COMMON MISTAKE

A region is a data centre.

A region is a group of AZs, and each AZ is one or more data centres.

Why the home region is ap-south-1 (Mumbai)

3

AZs in Mumbai
at the time of writing

1,550 km

Colombo to Mumbai
great-circle distance

23 ms

Estimated round trip
light in fibre, x 1.5 path

Region	AZs	Enabled by default	Distance	Est. round trip
Asia Pacific (Mumbai)	3	Yes	1,550 km	23 ms
Asia Pacific (Hyderabad)	3	No, opt-in	1,170 km	18 ms
Asia Pacific (Singapore)	3	Yes	2,730 km	41 ms
US East (N. Virginia)	6	Yes	14,400 km	216 ms

You have built this before

Every lab in this module runs in ap-south-1. Check the region before you build.

YOU ALREADY KNOW THIS

Two data centres in different suburbs on different power feeds, joined by dark fibre.

T

THARA HOSPITALS

Colombo had one basement and one flood took everything. The pilot lives in three AZs in Mumbai.

CONCEPT 4

Virtualisation and containerisation

Should the portal be a VM image or a container image?

40 min

Hypervisors and the Nitro System



- A **Type 1** hypervisor runs on the hardware; a **Type 2** runs as an application on a host OS
- Hardware-assisted virtualisation lets the CPU switch safely between VMs
- Every VM carries its own operating system and kernel
- On EC2, **Nitro** cards offload network, storage and security, leaving a very thin hypervisor

Containers share one kernel



- A container is a group of ordinary processes given a restricted view by the host kernel
- **Namespaces** give it its own process ids, network, mounts and hostname
- **Cgroups** limit the CPU and memory it may use
- No second kernel boots, so containers start fast and pack densely

YOU ALREADY KNOW THIS

A VLAN isolates at layer 2, a hypervisor isolates hardware, a container isolates a process tree.

Virtual machine or container?

Virtual machine

- **Boundary:** the hypervisor, separate kernels
- **Density:** fewer per host, each carries an OS
- **Start time:** seconds to minutes
- **Attack surface:** an escape must break the hypervisor

Container

- **Boundary:** namespaces and cgroups in one kernel
- **Density:** many more per host
- **Start time:** under a second to a few seconds
- **Attack surface:** an escape must break the shared kernel

A **different** boundary, not a thinner one.

Where each fits on AWS

- An EC2 instance is a VM
- A container runs inside a VM, or on a managed container service



Tomorrow you run the Thara pilot page natively and in a container on the same instance, and compare them.

COMMON MISTAKE

Containers are lightweight VMs.

They share the host kernel; the boundary is in a different place.

CONCEPT 5



Who secures what

Who can read patient data? The answer differs at each layer.

20 min



DATA PROTECTION OFFICER

Who can read patient data? I need an answer for every layer, not just "it is in the cloud".

The Data Protection Officer, appointed under Sri Lanka's Personal Data Protection Act.

Security of the cloud, security in the cloud

- AWS secures **of** the cloud: buildings, hardware, network, hypervisor
- Thara secures **in** the cloud: access, data, configuration
- EC2: Thara patches the OS. Managed database: AWS patches the engine

YOU ALREADY KNOW THIS

The ISP owns the fibre; you own the firewall rules behind it.

COMMON MISTAKE

The cloud is secure.

AWS secures the cloud; Thara secures what it puts in it.

Demonstration: console tour with a deliberate mistake



1. Sign in as an IAM user; find the region selector and the Explore AWS widget
2. Switch to us-east-1, create nothing, switch back. Where would a resource made there be?
3. Open IAM and look for its region selector
4. Open the Free Tier page and the credit balance

Activity: prior-knowledge diagnostic, 20 minutes



1. Subnetting: from 10.0.0.0/16 carve four /24s and name the broadcast of one (ten items)
2. Read a routing table and pick the route each packet takes (five items)
3. Shell: list, change directory, edit, view a log (five items)
4. Swap papers; we mark together now

Common mistakes



COMMON MISTAKE

A region is a data centre.

A region is a group of AZs, each one or more data centres.

COMMON MISTAKE

Containers are lightweight VMs.

They share the host kernel; the boundary is different.

COMMON MISTAKE

The cloud is secure.

Security of the cloud is AWS's; security in it is Thara's.

Before the next session



READ

Erl and Barceló Monroy

Chapters 3 and 4, plus Cloud Practitioner Essentials module 1 as recap.

SELF-CHECK

CLF-C02 bank

Domain 1 set A, 15 recall items.

PRACTICE

PQ1

A VM image or a container image for the Thara portal?

Posted in the practice folder.

Questions & Discussion



This afternoon: lock the account down before anything is built.



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Revision 47c64ea